



Fruit Measure App

User manual

Version 1.0.3



Universitat
de Lleida

IRTA^R

Institut
de Recerca i Tecnologia
Agroalimentàries



Generalitat de Catalunya
**Departament d'Agricultura,
Ramaderia, Pesca i Alimentació**



**Cofinançat per
la Unió Europea**

Credits

Fruit MeasureApp has been developed by the University of Lleida (UdL) and the Institute of Agro-Food Research and Technology (IRTA) within the framework of the demonstration activity “FruitMeasureApp: Validation and prototyping of an AI-based mobile application for measuring fruits in the field” (Activity co-financed by the EU through intervention 7201 of the CAP Strategic Plan 2023-2027).

Complementary resources related to FruitMeasureApp are available on the website:

<https://fruitmeasureapp.udl.cat/>



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YOLO License

This application incorporates components of Ultralytics YOLOv11.

YOLOv11 © Ultralytics and contributors. Licensed under the GNU Affero General Public License v3.0 (AGPL-3.0).

This application is distributed without any warranty.

Source code available at:

<https://github.com/GRAP-UdL-AT/FruitMeasureApp>

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Introduction

FruitMeasureApp is a mobile application based on artificial intelligence models developed by the University of Lleida (UdL) and the Institute of Agro-Food Research and Technology (IRTA) that allows users to accurately and quickly measure apples on the tree. The models on which FruitMeasureApp is based have been trained with apples with diameters between 10 mm and 100 mm.

With FruitMeasureApp you can view fruit data, track growth and compare results between different dates or plots, in an orderly and clear way. Simply take a photo with your mobile phone attached to the [FruitMeasureStick](#) and FruitMeasureApp will handle all the processing automatically.

FruitMeasureApp is designed to measure apples in the range of 10 mm to 100 mm and is not intended for use with any other variety of fruit or outside the specified range.

Disclaimer

The values provided by this application are indicative estimates and should not be interpreted as professional or technical advice, nor used as the sole basis for significant decisions. This tool does not replace certified measuring instruments or the judgment of a qualified professional.

Although developed with the utmost care, the accuracy, completeness, or suitability of the results under all circumstances is not guaranteed. The interpretation and use of the information obtained are the sole responsibility of the user.

The development team shall not, under any circumstances, be held liable for any direct or indirect damages arising from the use, misuse, or interpretation of the data generated by the application.

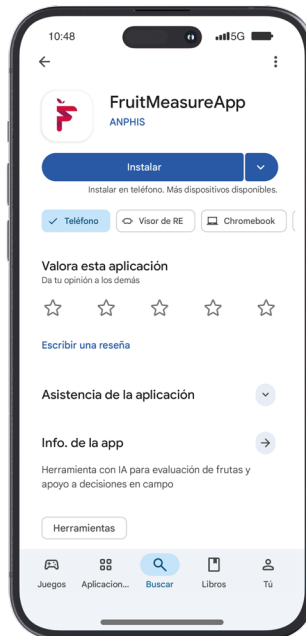
Use of this application implies express acceptance of these terms.

Before we start

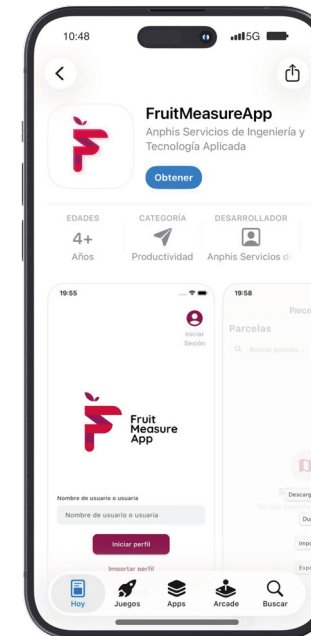
a. Download FruitMeasureApp

Click on one of the following links depending on the app store of your mobile device or tablet.

Play Store (Android devices)



App Store (iOS devices)



If not, in your app store search engine, search for FruitMeasureApp and select the application. Click the install button and wait for the download to complete.

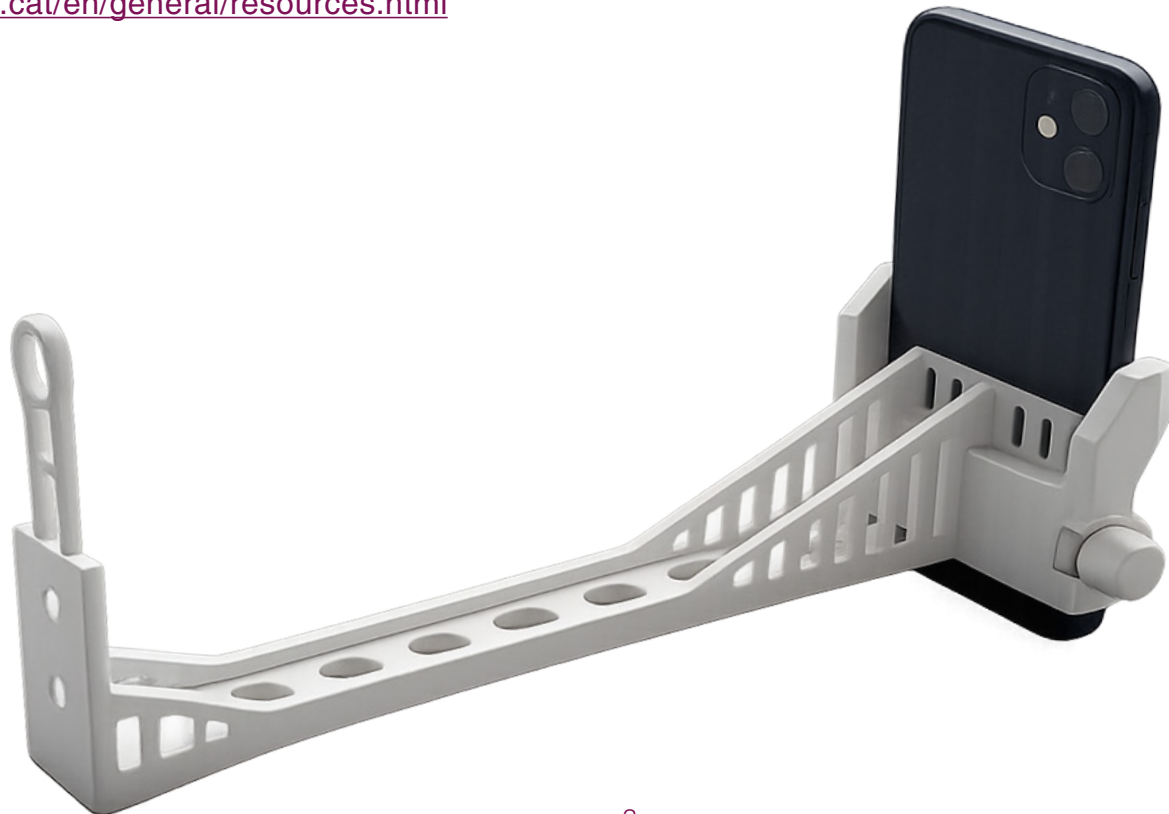
Before we start

b. FruitMeasureStick

“FruitMeasureApp requires the FruitMeasureStick. This accessory attaches to the smartphone and provides a reference scale, enabling accurate estimation of apple size.

The FruitMeasureStick has been designed to be printed with a 3D printer. To download the file, see how to assemble it, or the different printing service options, access this link:

<https://fruitmeasureapp.udl.cat/en/general/resources.html>



Before we start

c. Model types

Cluster



Cluster is the model used for fruits in the early stages of growth that appear grouped in a cluster. In this mode, the application measures all the fruits in the corymb in a single capture.

Fruit



Fruit is the model used when the apple is in advanced stages of growth and no longer appears grouped in a corymb. In this mode the application only measures the fruit where the support is placed.

It is important to choose the right model to use the application optimally.

Before we start

d. How to take the photographs

To use FruitMeasureApp, the images or photographs you use must be in 3:4 vertical format. From the camera you can adjust the aspect ratio of the photographs that are taken.

When taking the photograph, it is important to take into account the following:

1. If the image is for the Cluster model, the circular part of the FruitMeasureStick must be located in the middle plane of the corymb.
2. On the other hand, if it is for the Fruit model, the circular part of the FruitMeasureStick must touch the surface of the apple



Before we start

e. Camera support distance

This is the distance between the end of the [FruitMeasureStick](#) and the camera of the mobile device. This parameter is important for the application to make an optical correction of the measurements made with the “fruit” mode. The standard [FruitMeasureStick](#) has a length of 250 mm. If you do not use the standard [FruitMeasureStick](#) and it is of a different length, you will need to modify this parameter so that the app makes correct measurements.

The distance from the support to the camera can be modified from [Edit profile](#).

f. Confidence level

The confidence level is the certainty that the application has that the detected object is really a fruit. This parameter allows you to discard the measurements of those detections with a confidence level lower than the configured threshold.

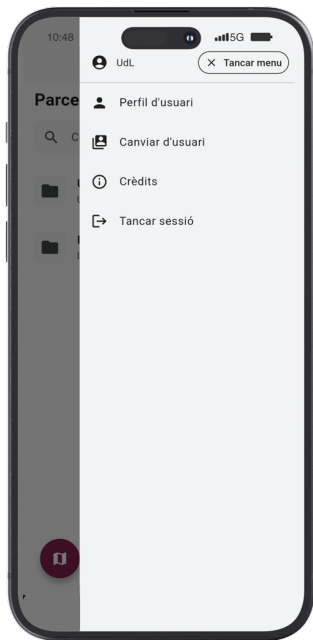
If a high confidence level is set, fewer apples will be detected, but with more certainty (fewer false positives). On the other hand, with a low confidence level, more apples will be detected, but there is a risk that more false positives will appear.

The confidence level can be modified from [Edit profile](#).

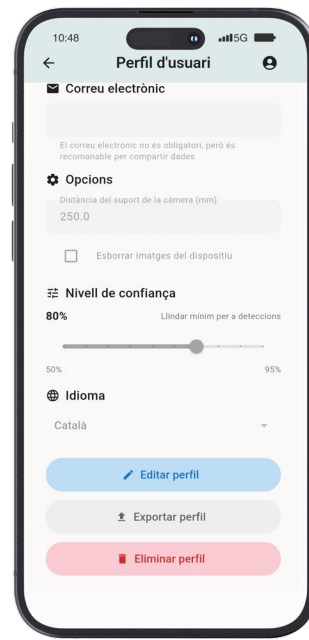
Before we start

g. Delete images from the device

The application stores the images with the measurements on your mobile device. In case you do not want to save these images, you can activate this function.



1. Select  and click **User Profile**.



2. Scroll down to the bottom of the page and click **Edit Profile**.



3. Find the **Delete images from device** option and check the box.



4. Go back to the bottom of the page and click **Save Changes**.

Nomenclature

a. What is a plot?

A plot is a folder within the application where the measurements of a single plot are saved.

b. What is a measurement day?

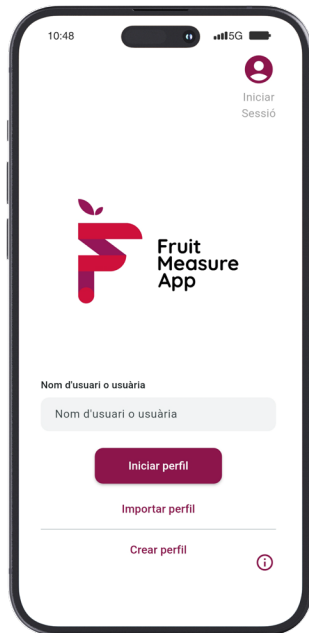
A measurement day is a folder within your plot where information for one day is stored. Within a plot folder there can be multiple measurement folders.

c. What is a measurement record?

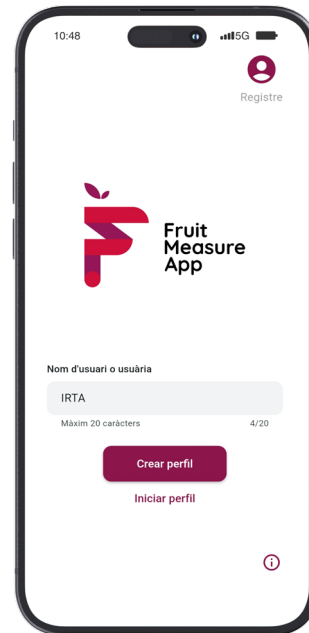
A measurement record consists of information about a fruit or a corymb. A measurement folder can contain multiple measurement records.

Access or login

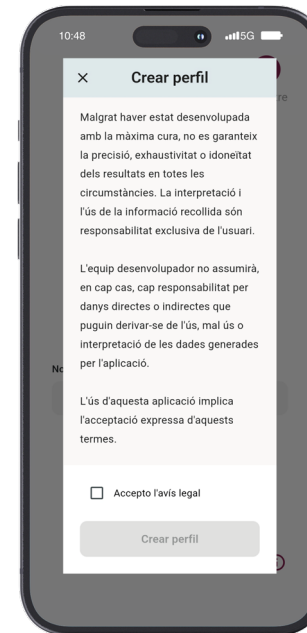
a. Create a profile



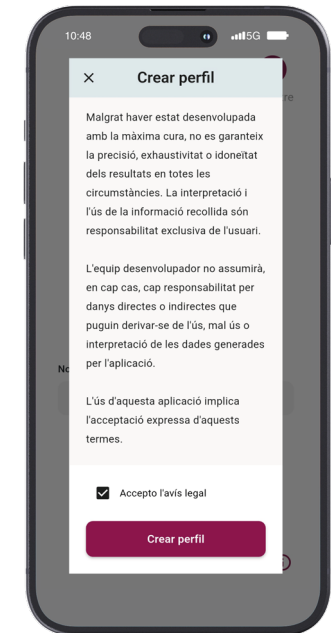
1. On the home screen, select **Create Profile**.



2. Enter the username you want to assign.



3. Click **Create Profile**, read the legal notice and check the acceptance box.

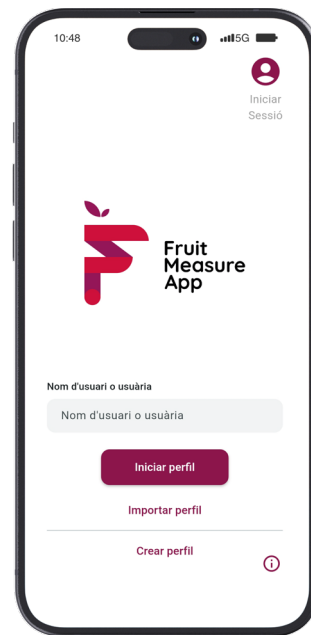


4. Finally, click **Create Profile** again.

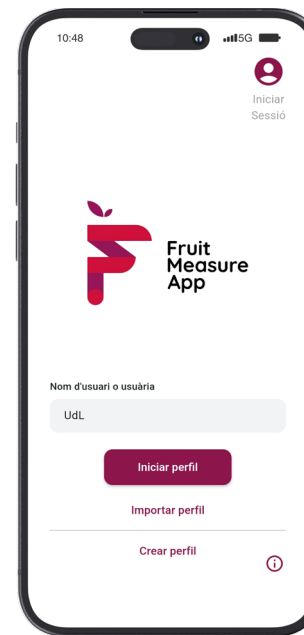
All user profiles are stored on the same device to make it easier to share the device.

Access or login

b. Login



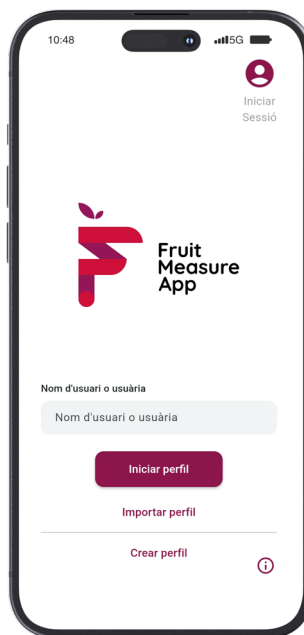
1. On the initial screen, type the name of the profile you want to start.



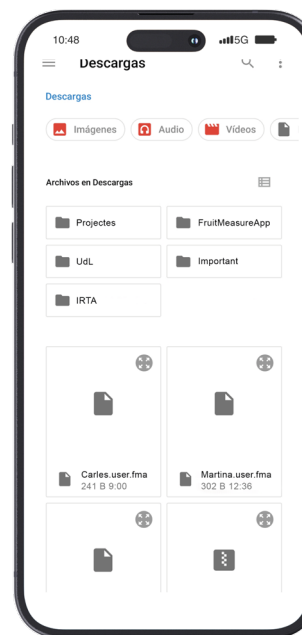
2. Finish by clicking **Start Profile**.

Access or login

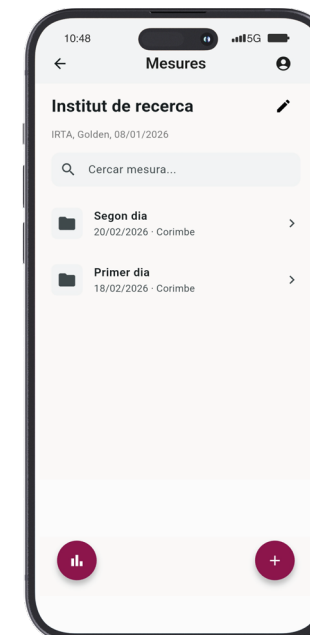
c. Import a profile



1. On the home screen, select **Import Profile**.



2. The file folder will open, find the desired file (*example: carles.user.fma*) and select it.

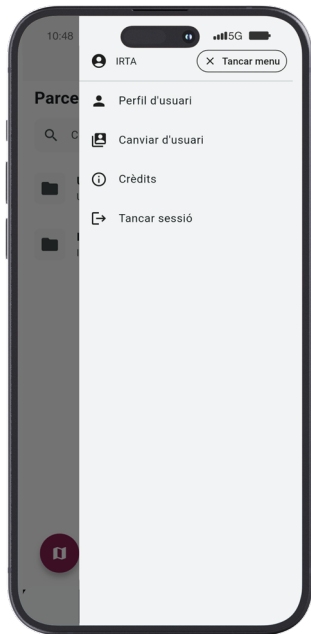


3. The profile will open directly in the app.

This feature allows you to import user data from another device.

User profile

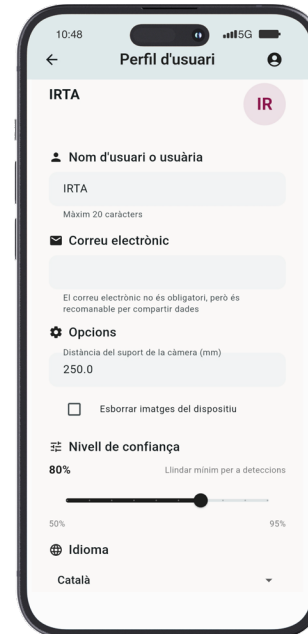
a. Edit a profile



1. Select  and **User Profile**.



2. Scroll down to the bottom and click **Edit Profile**.



3. You can modify data such as name, email, camera support distance, whether or not to save photos on the device, confidence level or language.



4. To save, at the bottom of the page click **Save changes**.

User profile

b. Export a profile



1. Select  and **User Profile**.



2. Scroll down to the bottom and click **Export Profile**. You can rename the file if you want.

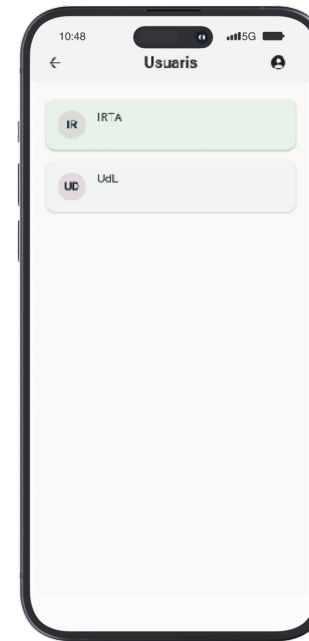
It is important to keep the .user.fma, so that you can later import it correctly (*example: carles.user.fma*).

User profile

c. Switch user



1. Select  and **Switch user**.



2. A page will appear with all available users, choose the desired profile.

User profile

d. Log out



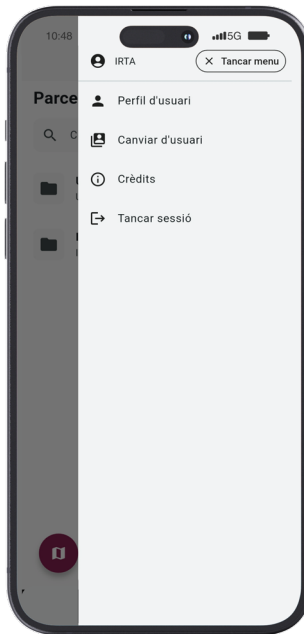
1. Select  and **Log Out**.



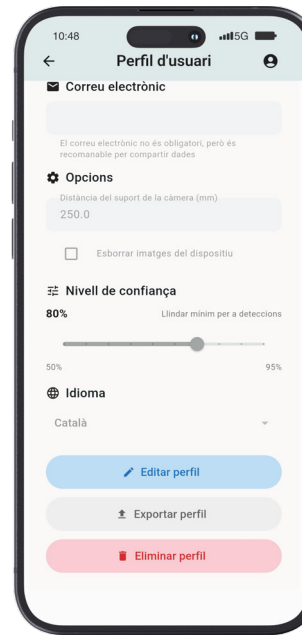
2. Finally, confirm the action.

User profile

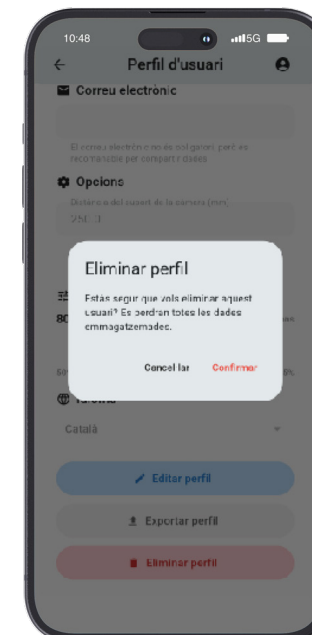
e. Delete a profile



1. Select  and **User Profile**.



2. Scroll down to the bottom and press **Delete Profile**.

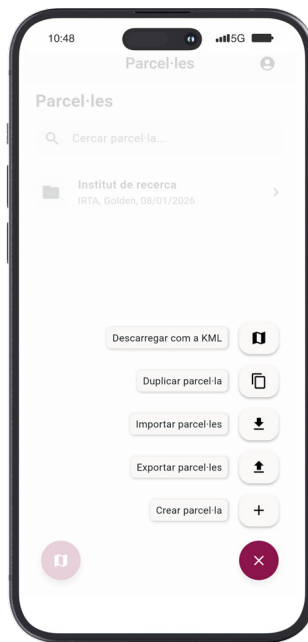


3. Finally, confirm the action.

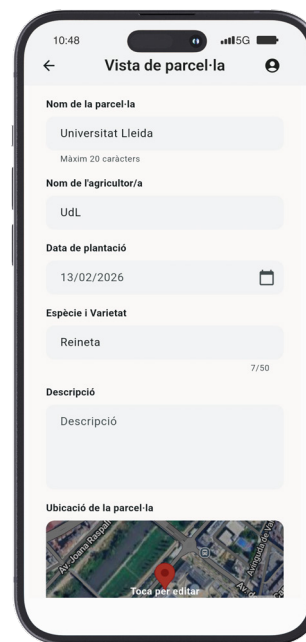
Once a profile is deleted it cannot be recovered.

Plots

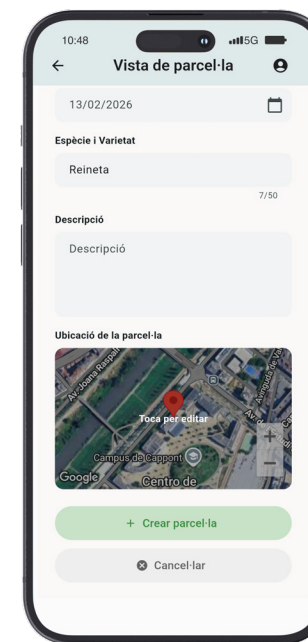
a. Create a plot



1. Click  and select **Create plot**.



2. Fill in the information indicated: the plot and farmer's name, the date, the specie, the description and the location.

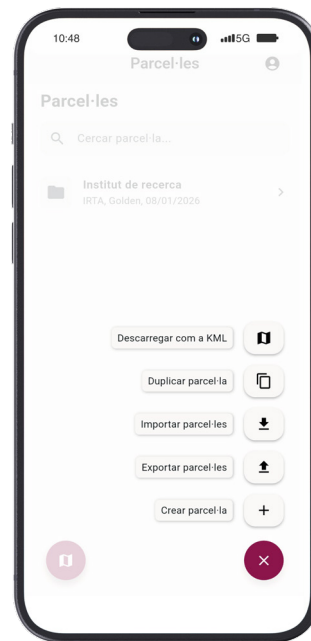


3. When finished, click the **Create plot** button to complete the operation.

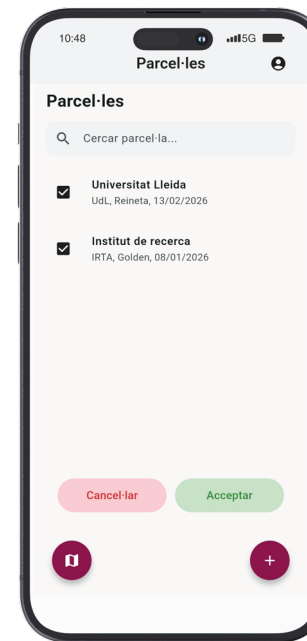
To view the plots on a map edit the location.

Plots

b. Export a plot



1. Click  and select **Export plots**.

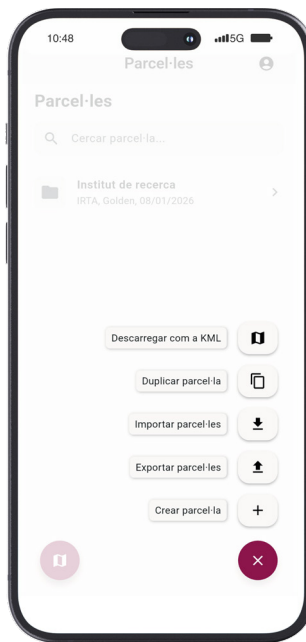


2. Select the plots you wish to export and click **Accept** to continue. You can change the file name if you wish.

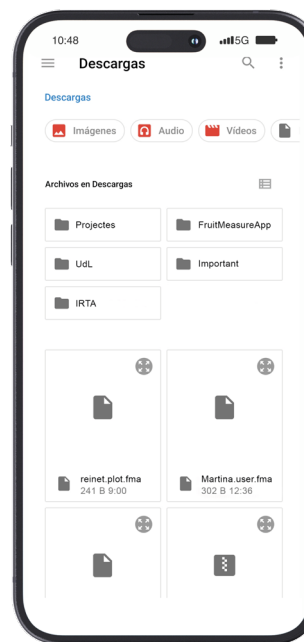
It is important to keep the .plot.fma, so that later you can import it correctly (*example: reineta.plot.fma*).

Plots

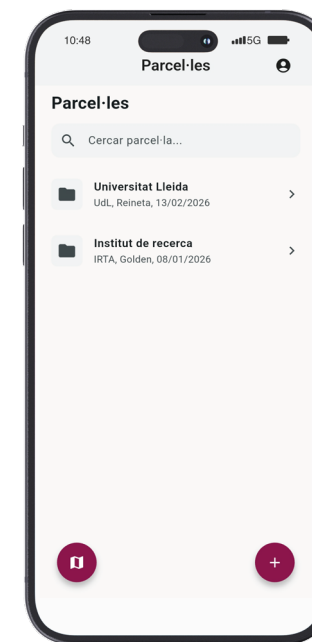
c. Import a plot



1. Click  and select **Import plots**.



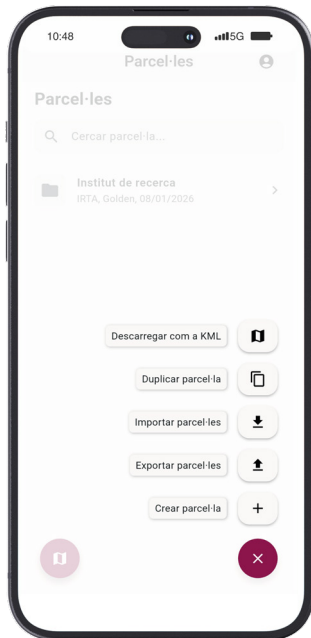
2. The file folder will open, find the desired file (*example: reineta.plot.fma*) and select it.



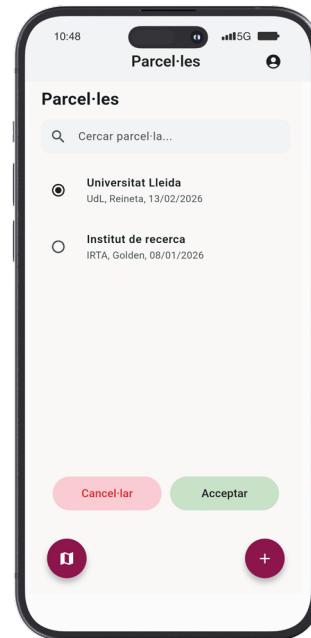
3. You will see the imported plot in the corresponding menu.

Plots

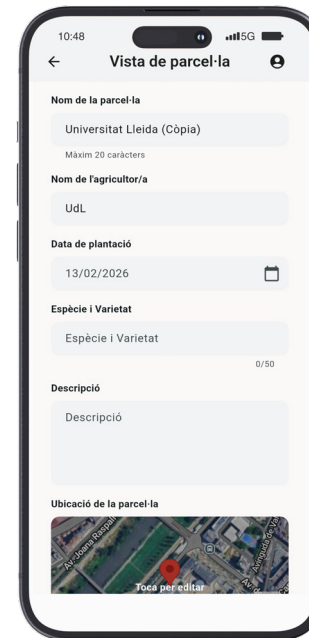
d. Duplicate a plot



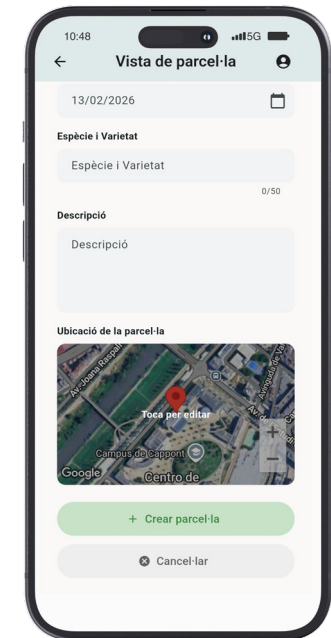
1. Click **+** and select **Duplicate plot**.



2. Mark the plot you want to duplicate and click **Accept**.



3. You can modify the following information: the plot and farmer's name, the date, the specie, the description and the location.

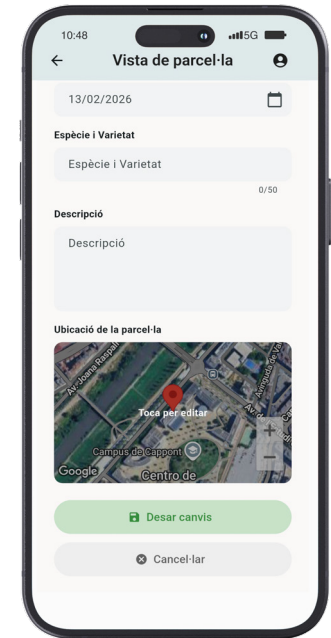
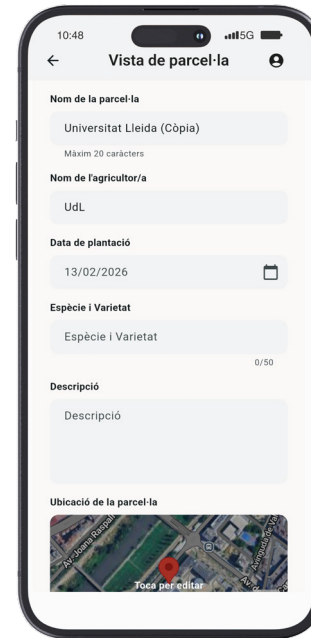
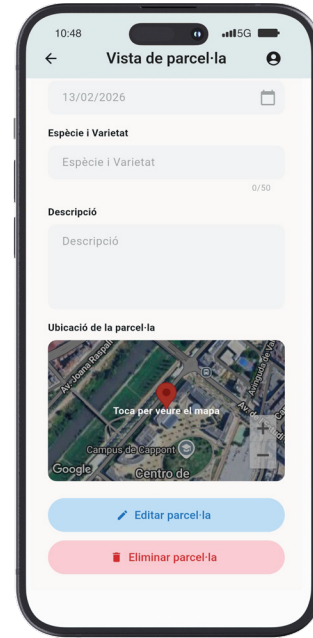
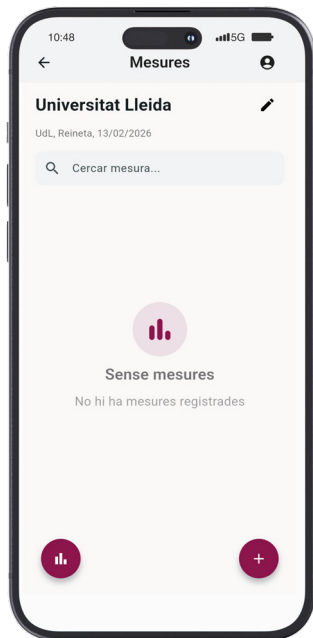


4. Finally, click **Create plot**.

Duplicate plots maintain all the measurements of the original plot.

Plots

e. Edit a plot

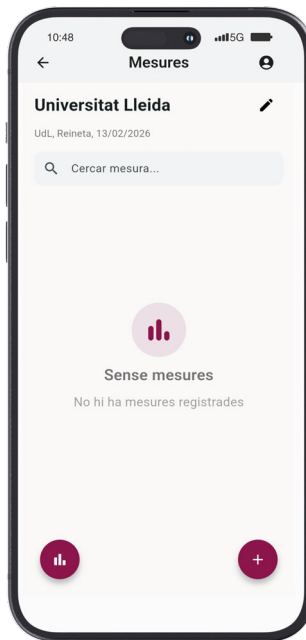


1. Select the plot you want to modify. Once inside, click .
2. Scroll down to the bottom and click **Edit plot**.
3. You can modify the following information: the plot and farmer's name, the date, the specie, the description and the location.
4. After, click **Save changes**.

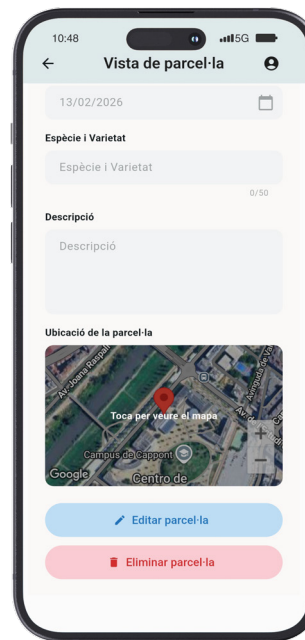
To view the plots on a map edit the location.

Plots

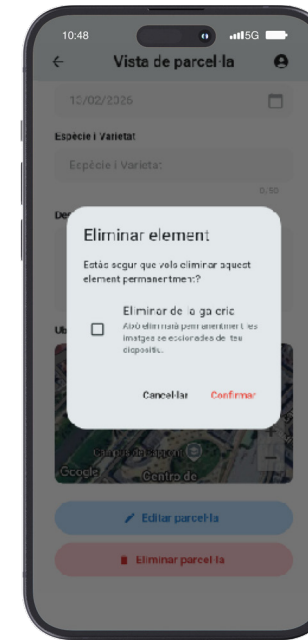
f. Delete a plot



1. Select the plot you want to delete. Once inside, click .



2. Scroll down to the bottom and click **Delete plot**.

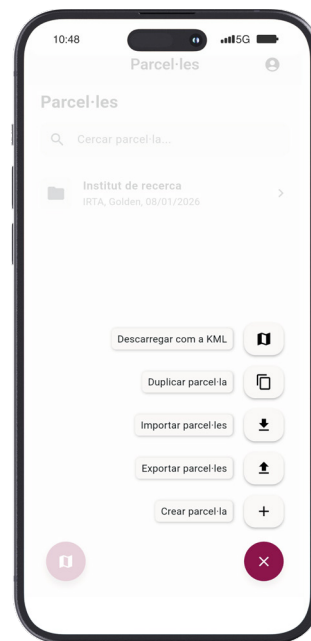


3. A message appears asking if you want to delete the images from the gallery. If so, check the box and confirm.

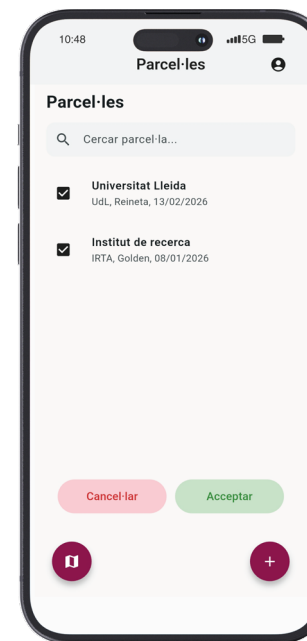
A deleted plot cannot be recovered.

Plots

g. Download the plot as a KML



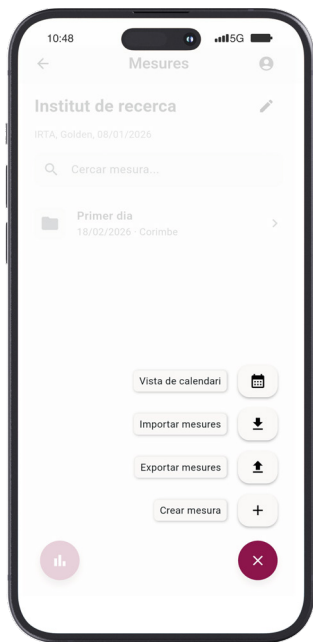
1. Click  and select **Download as KML**.



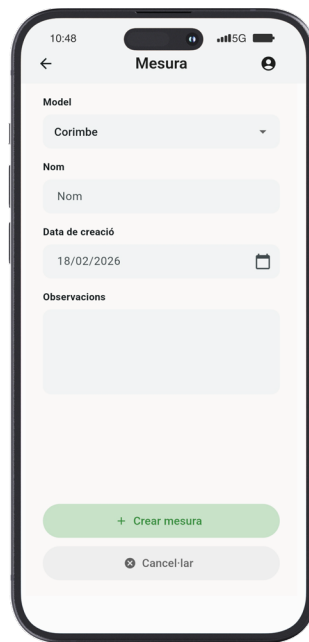
2. Check the plot you want to export in this format and accept. You can change the file name if you want.

Measures

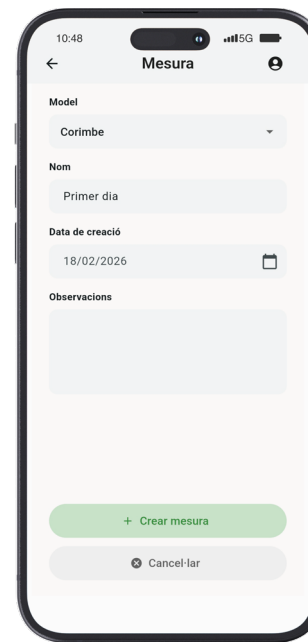
a. Create a measurement day



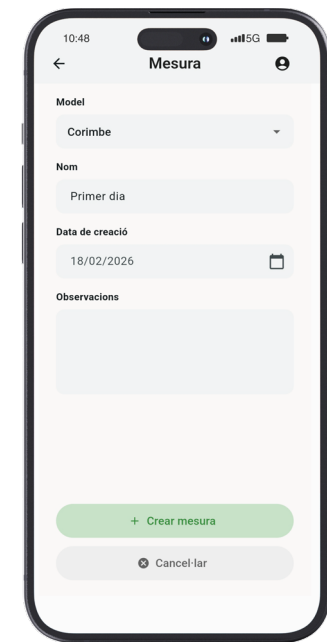
1. Within the plot where you want to add a measure, click **+** and select **Create measure**.



2. Select, within the Model field, one of the following options: Fruit or Cluster.



3. Once the model is selected, you can modify other optional information, such as the name, creation date or observations.

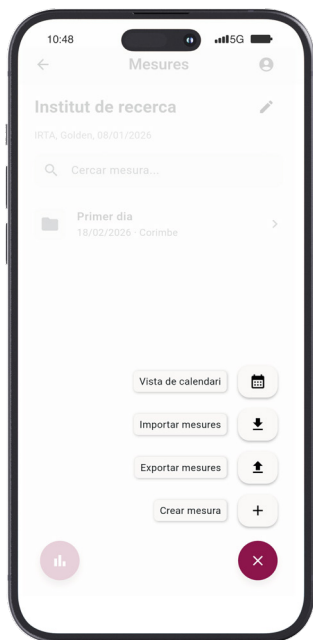


4. To finish, click **Create measure**.

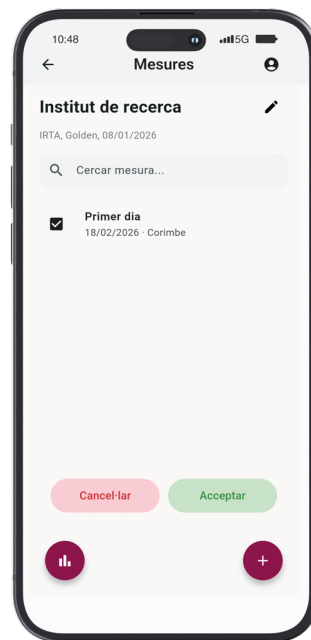
It is important to choose the correct model to make optimal use of the application.

Measures

b. Export a measurement day



1. Within the corresponding plot, click **+** and select **Export measurements**.



2. Mark one or more measurements that you want to export and click **Accept**.

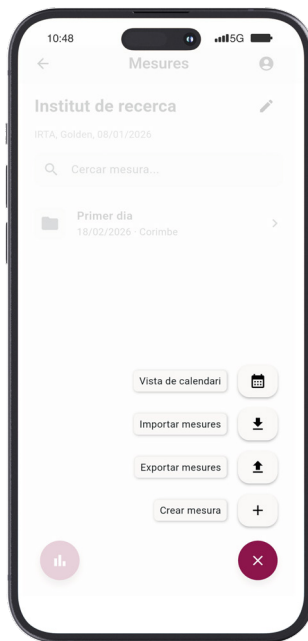


3. Select the type of file format in which you want to export: JSON, CSV or TXT. Once chosen, you can change the name of the file.

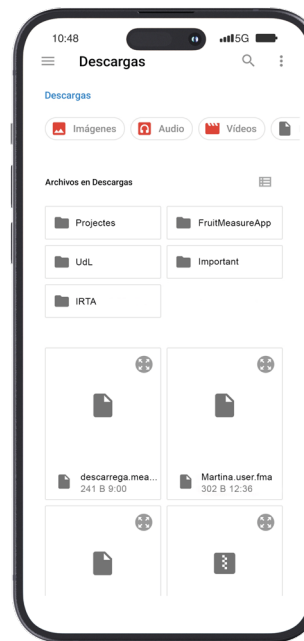
- **JSON:** Select this format if you want to **import the measurements into the app**. The `.measurements.fma` extension must be kept to be able to import them correctly.
- **CSV:** Allows you to view the data in a table format, such as Excel or similar.
- **TXT:** Allows you to view the data in a plain text format.

Measures

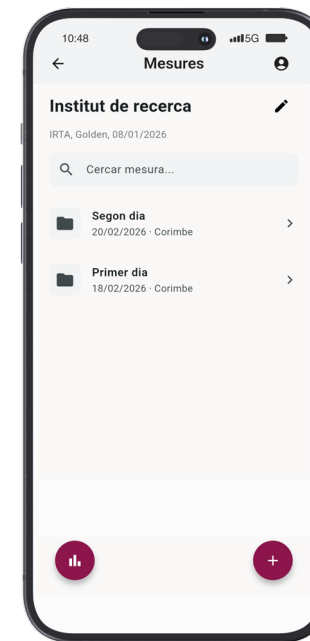
c. Import a measurement day



1. Within the corresponding plot, click  and select Import measurements.



2. Find your JSON file .measurements.fma (example: download.measurements.fma) and select it.



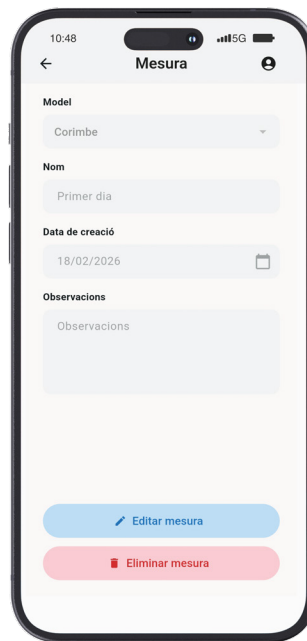
3. You will see the imported measurement in the measurements folder.

Measures

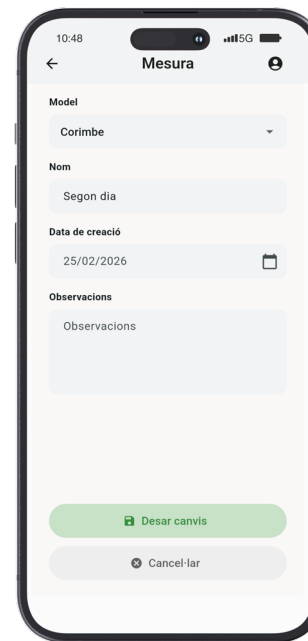
d. Edit a measurement day



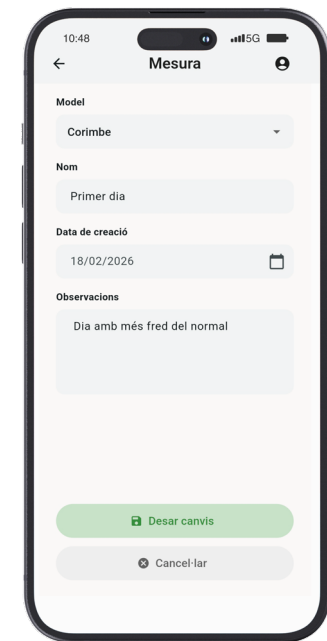
1. Within the measure you want to modify, click .



2. Click **Edit Measure**.



3. Change the relevant information such as the Model, name, creation date, or observations.

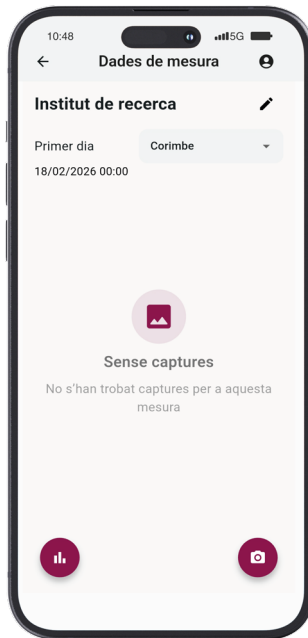


4. To finish, click the **Save Changes** button.

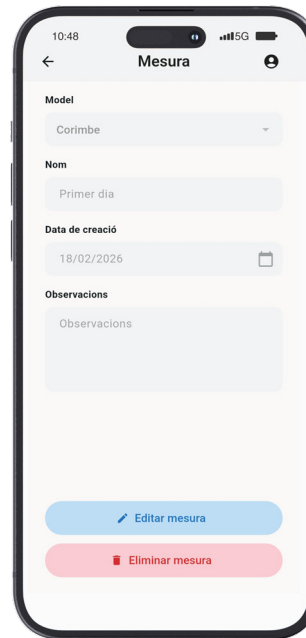
It is important to choose the correct model to make optimal use of the application.

Measures

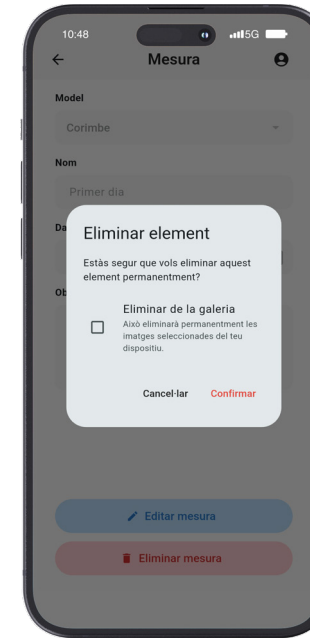
e. Delete a measurement day



1. Within the measurement day you want to delete, click .



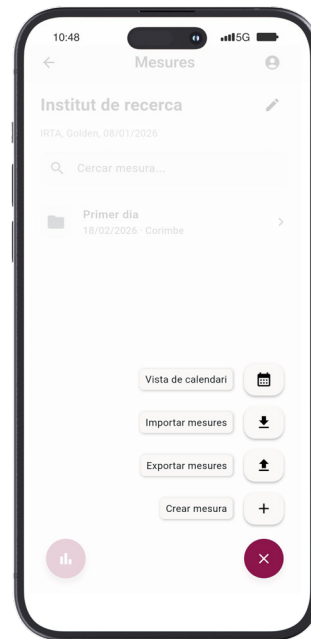
2. Press **Delete measurement**.



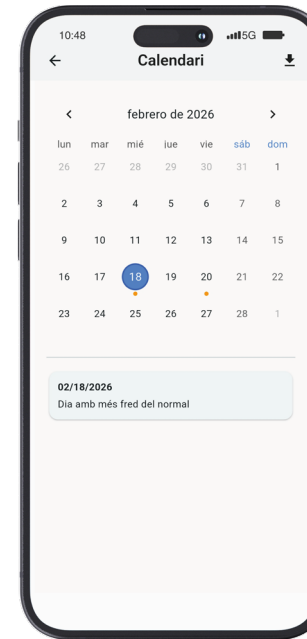
3. A message appears asking if you want to also delete the images from the gallery. If so, check the box and confirm.

Calendar

The calendar is used to view the days on which a measurement has been added to the plot. It also allows you to check the number of measurements taken each day and the associated observations.



1. To access it, click  within the corresponding plot and select **Calendar view**.



2. To view the measurements and observations for a specific day, click on the corresponding day. The information will be displayed at the bottom of the screen.

Measurement data

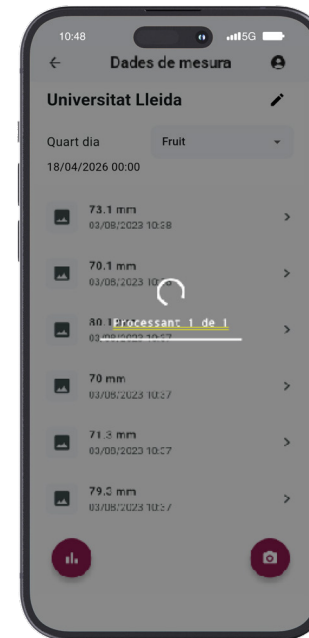
a. Add data from the gallery



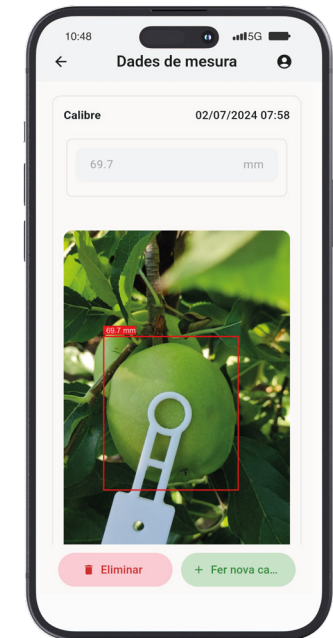
1. Within the corresponding measurement day, select  and click **Select image from gallery**.



2. The application will open the gallery, here you can select one or more photos and then confirm the selection.



3. The app will process the image and open a new tab with the fruit information. (if you select more than one image the measurement menu will open).

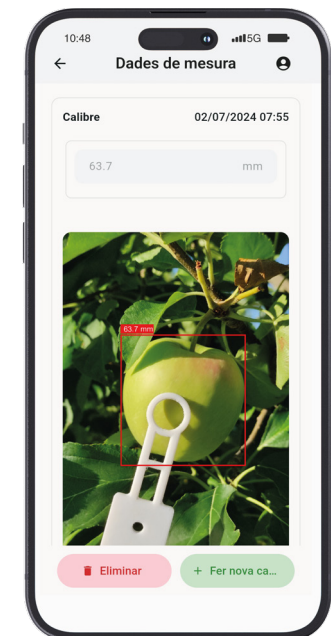
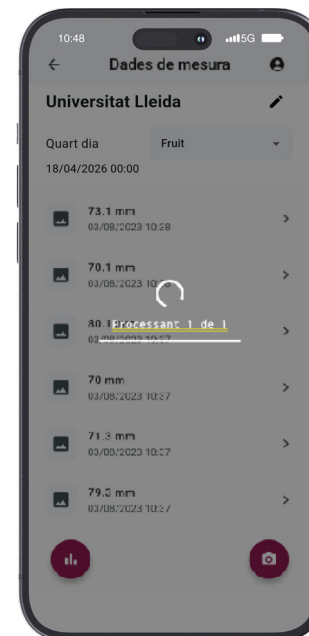


4. Select **Take new capture** to add more images or press  to save.

To view the location of the data, you must allow location in both the application and the mobile device.

Measurement data

b. Add data from the camera

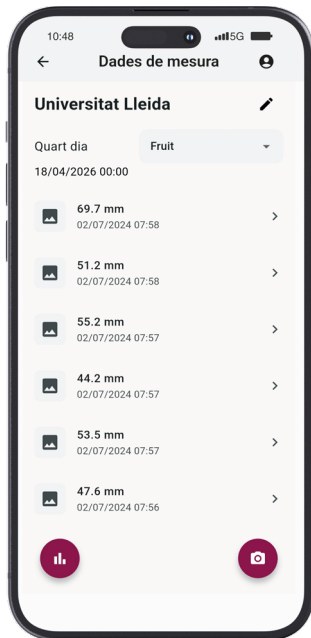


1. Within the corresponding measurement day, select  and click *Take capture*.
2. The application will open the camera, take the photo with the FruitMeasureStick.
3. The application will process the image and open a new page with the fruit information.
4. Select **Take new** capture to continue or press  to save.

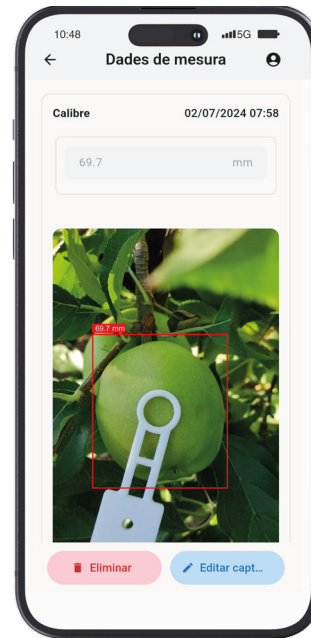
To view the location of the data, you must allow location in both the application and the mobile device.

Measurement data

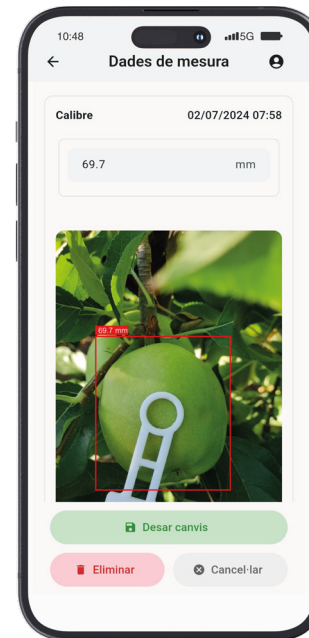
c. Edit a capture



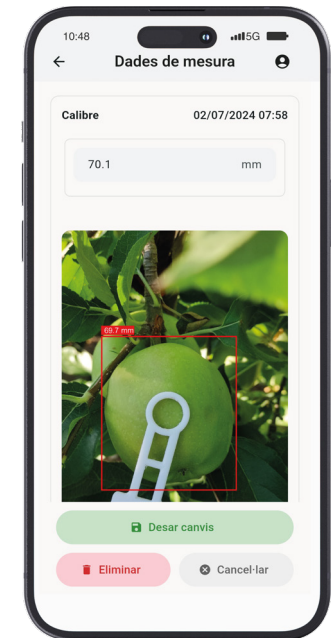
1. Select the capture you want to modify.



2. The data information will open, click the **Edit shot** button.



3. You can modify the measurements, caliber and location.

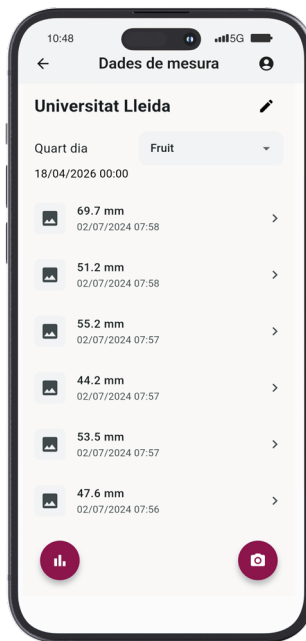


4. To finish, click **Save changes**.

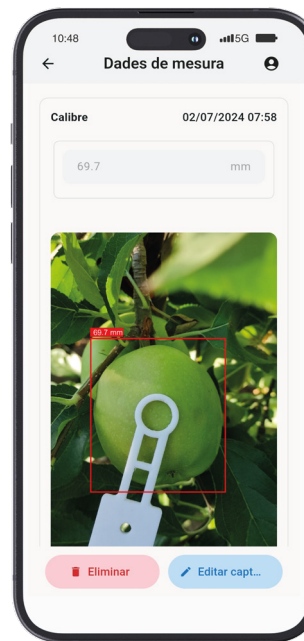
To view the location of the data, you must allow location in both the application and the mobile device.

Measurement data

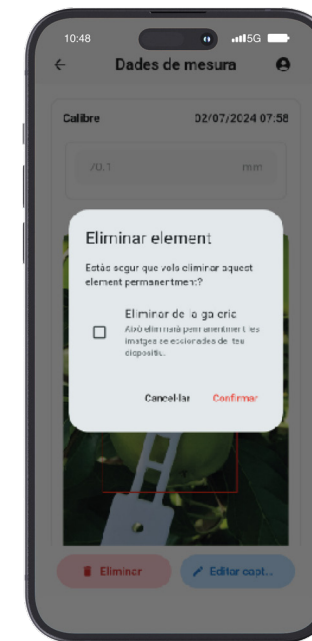
d. Delete a capture



1. Select the capture you wish to delete.



2. Then, click the **Delete** button.

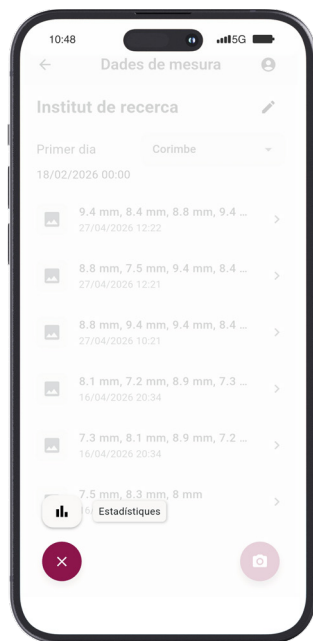


3. A message appears asking if you want to delete the images from the gallery. If so, check the box and confirm.

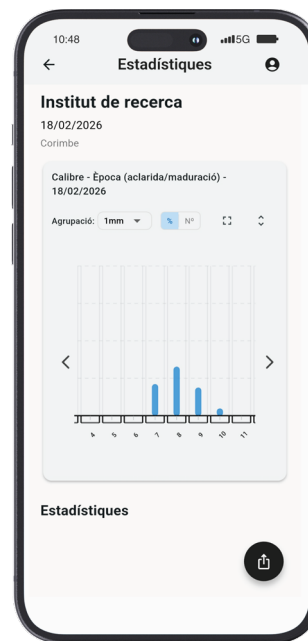
A deleted capture cannot be recovered.

Statistics and data comparison

a. Statistics of a measurement



1. Within the folder of the desired measurement day, select  and click **Statistics**.

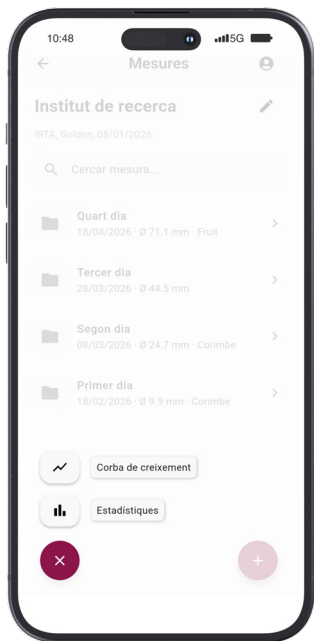


2. A bar graph appears with the distribution of the measured sizes. Statistical data on the minimum and maximum size, the number of fruits, images and fruits per image are also presented.

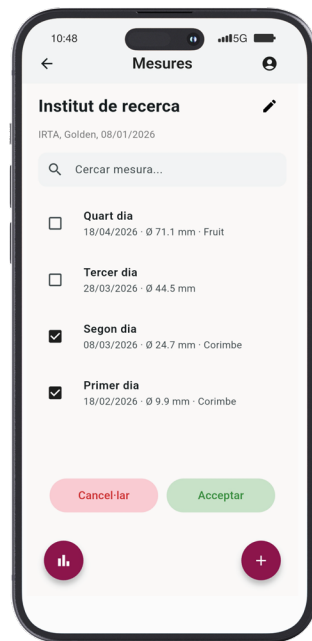
- Clicking on the **Grouping** option will open a drop-down menu that allows you to change the diameter range that each bar of the histogram represents.
- By clicking on % or N^o, you can select whether the number of fruits is shown as a percentage of the total or as an absolute number.
- By clicking on , the graph will be expanded by rotating the display 180° and inverting the axes to make it easier to read. To exit this view, press  or .
- By clicking on , the graph will automatically adjust to allow the display of all the bars of the histogram.

Statistics and data comparison

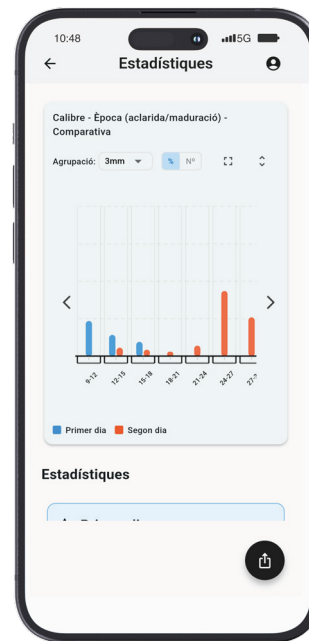
b. Data comparison



1. Within the folder of the desired plot, select  and click on **Statistics**.



2. Select the measures days you want to compare (up to a maximum of 6) and click **Accept**.



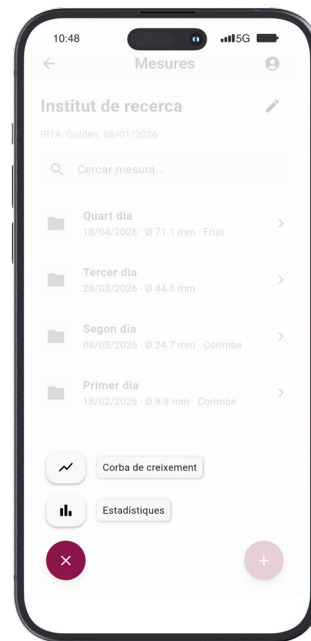
3. The size distributions are compared and the statistical data for each day of measurement is displayed.

- Clicking on the **Grouping** option will open a drop-down menu that allows you to change the diameter range that each bar of the histogram represents.
- By clicking on % or N°, you can select whether the number of fruits is shown as a percentage of the total or as an absolute number.
- By clicking on , the graph will be expanded by rotating the display 180° and inverting the axes to make it easier to read. To exit this view, press  or .
- By clicking on , the graph will automatically adjust to allow the display of all the bars of the histogram.

Each measurement day is represented by a different color, with the legend located under the same graph.

Statistics and data comparison

c. Growth curve



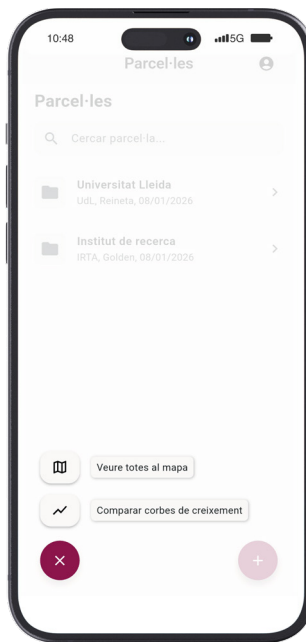
1. From a plot folder select  and click on **Growth Curve**.



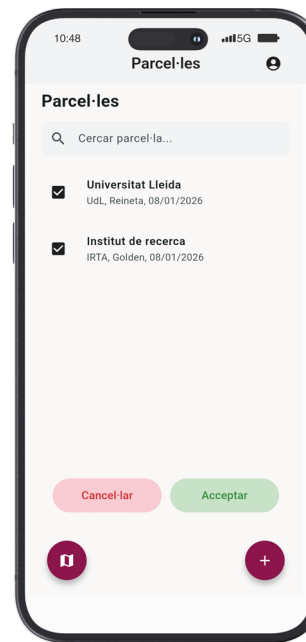
2. A graph appears showing the temporal evolution of the plot measurements (the average diameter of each day) and its growth rate.

Statistics and data comparison

d. Compare growth curves



1. From the Plots menu, select  and click **Compare growth curve**.



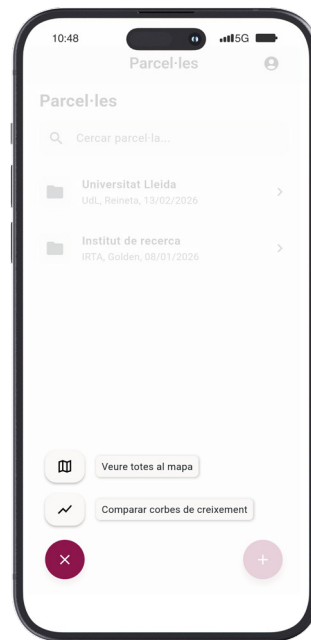
2. Select the plots you want to compare up to a maximum of 6 and click **Accept**.



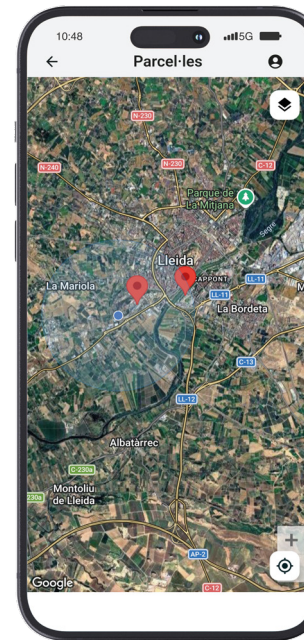
3. A graph appears showing the temporal evolution of the plots (the average diameter of each day) and the current growth rates.

Each plot is represented by a different color, with the legend located under the same graph.

Viewing plots on a map



1. From Plots, select  and click **View all on map**.



2. A map will open showing all plots with red icons. Clicking on them will show the name and variety of the corresponding plot.

To be able to view the plots on a map, you must have defined their location when creating them, or edit them later.



Fruit
Measure
App